SET B

Register No	RA2512044015108

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
21CSC555J – MACHINE LEARNING ALGORITHMS
CYCLE TEST – I
Theory

Program Offered: M. Tech
Max. Marks: 40

Year / Semester: I/ I
Duration: 1.5 Hrs
Date of Exam: 27/09/25

Part A (2 x 20 = 40 marks)

Answer any two questions

1. (i). Explain simple linear regression using an example dataset, along with a detailed breakdown of the mathematical calculations involved in deriving the equation? 10

   (ii)What are the evaluation metrics that are used for regression in Machine Learning? explain Mean Absolute Error (MAE) and Mean Squared Error (MSE)? 5

   (iii). Explain the concept of Overfitting and underfitting? 5

2 (i). What is Logistic Regression? How you will convert the line equation into Sigmoid curve? 5

   (ii). How you will handle more than two class using Logistic regression? Explain with example? 10

   (iii). What is Regularization and explain different types of it? 5

3 (i). Explain in detail about Cross validation and its types? 10

   (ii). What are the different performance metrics of a Classification model? Explain Confusion matrix table with example? 5

   (iii). What is loss function, explain how Gradient descent helps to Minimize the loss 5